

Rongfeng Cheng | Medical AI Research

Guangzhou, China

✉ eb1287945452@mail.scut.edu.cn • 🌐 crf2004.github.io/en • 📞 0009-0002-3021-9647

Education

South China University of Technology (SCUT)

Guangzhou, China

Bachelor of Management in Big Data Management and Application

2023–2027

Cumulative GPA: 3.46/4.0; average score: 85.18/100 (through Semester 6).

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Internship Experience

AI Research Institute, Guangdong Second Provincial General Hospital

LLM Development Engineer Intern

Nov 2024–Present

- Developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

Guangzhou Yunzhan Zhichuang Information Technology Co., Ltd.

Backend & Algorithm Development Intern

Jul 2025–Jan 2026

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company's iTwins product.

Manuscripts & Publications

Under Review [1]: R. Cheng et al. “DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification.” *IEEE BIBM 2026*. First author.

Under Review [2]: R. Cheng et al. “A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification.” *IEEE JBHI*. First author.

Major Revision [3]: “Implicit Medical Ethical Deficits in Large Language Models: A Benchmarking and Alignment Study.” Submitted to *Nature Communications*. Sixth author; leading the revision-stage experiments for the response to reviewers.

Ongoing: PathCLIP: Evidence-Aware ECG–Text Representation Learning. Ongoing first-author study of ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

Apr 2026 [4]: J. He, S. Qin, H. Zou, F. Jing, W. Liu, C. Li, Q. Wang, Z. Hu, K. Song, Z. Xu, Q. Luo, Y. Cheng, R. Cheng, H. Zhang, and W. Cheng. “CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation.” *iScience*. doi:10.1016/j.isci.2026.115073.

Research & Project Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification

Jun 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman $r = 0.843$ across 108 points; $r = 0.892$ after source-target-label aggregation).

A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification

Jan 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions.

Co-founder & Core Developer

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

Myocardial Ischemia Knowledge Graph

2025

Project Lead, SCUT Student Research Program

Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

Honors & Awards

2025: Campus Gold Award, Internet+ Competition – Dingbei Xinxin cardiovascular health-management project.

2025: Second Prize, 9th Guangzhou Mathematical Modeling University League – fruit classification with ConvNeXt.

2025: Third Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Division – nonlinear programming, genetic algorithms, and particle-swarm optimization.

Patents & Software IP

Patent: Method, System, and Electronic Device for Medical Question-Answering Output Assurance – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

Patent: Method and System for Integrative Chinese–Western Medicine-Assisted Acne Assessment Based on Multimodal Data Fusion – First Inventor; application submitted Jul 2026; examination documents pending.

Software IP: Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

Skills

Modeling: PyTorch, XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

LLM & Knowledge Systems: LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

Evaluation & Data Curation: RAG/retrieval evaluation, perturbation analysis, external validation, schema design, data harmonization

Engineering: Python, TypeScript, SQL, Neo4j, Flask, Git

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528